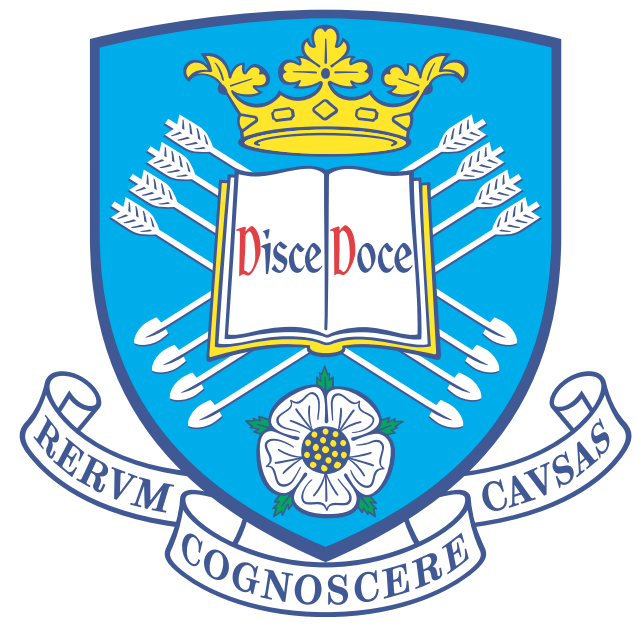




STRUCTURED SPARSE MODELLING WITH HIERARCHICAL GP

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Introduction

Structured sparsity naturally arises in many statistical models.

- Sparsity principle: signals are described with a small number of non-zero coefficients in special bases
- Structured sparsity principle: non-zero signal coefficients are grouped spatially or temporally forming a structure

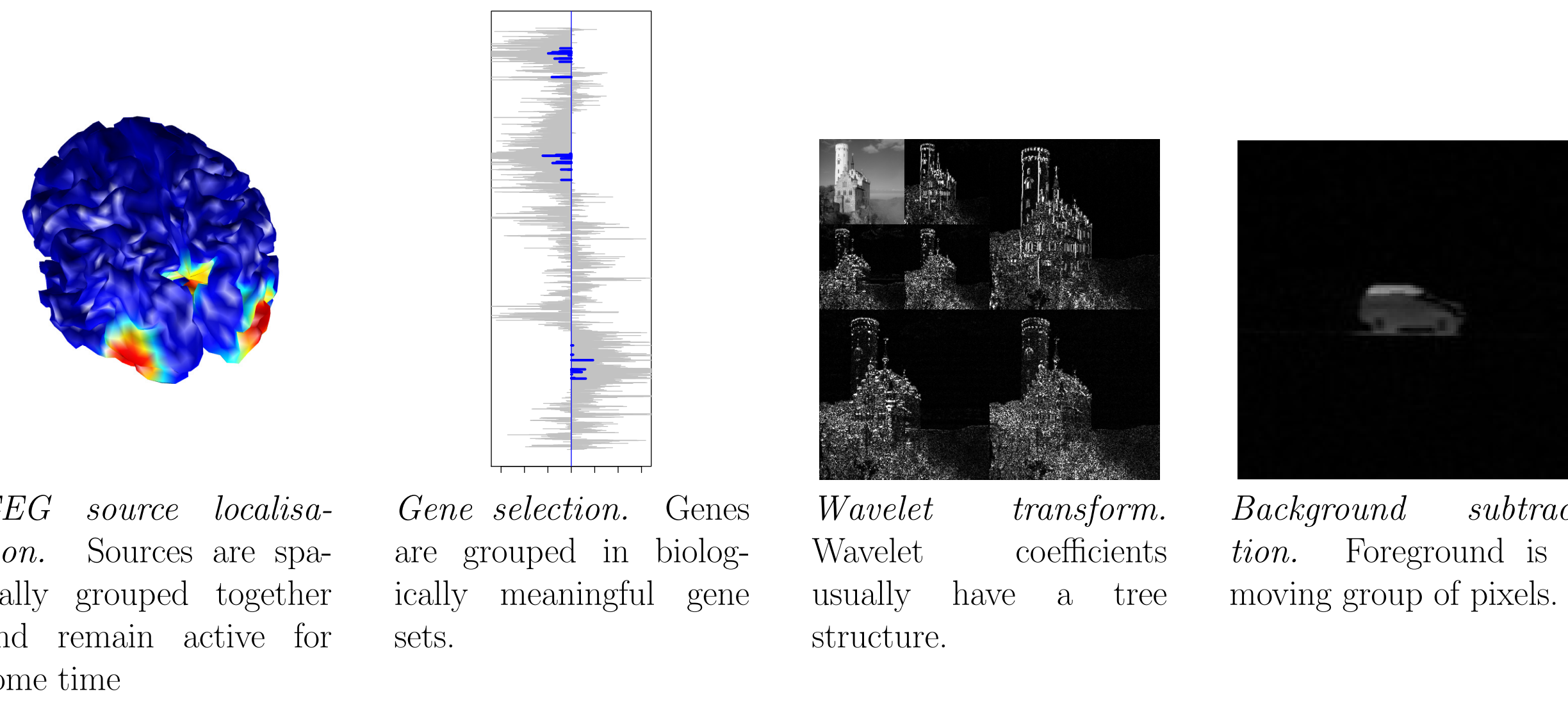


Fig. 1: Applications of structured sparse modelling

Spike and slab models

A Bayesian approach for sparse linear regression. The spike and slab prior induces a part of the variables being zeros.

Linear regression problem:

$$\begin{aligned} \mathbf{y}_t &= \mathbf{X}\boldsymbol{\beta}_t + \boldsymbol{\varepsilon}_t, \\ \boldsymbol{\varepsilon}_t &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \\ \mathbf{y}_t &\sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}_t, \sigma^2 \mathbf{I}). \end{aligned}$$

Spike and slab sparsity-promoting prior:

$$\begin{aligned} \beta_{it} &\sim \omega_{it}\delta_0 + (1 - \omega_{it})\mathcal{N}(0, \sigma_\beta^2), \\ \omega_{it} &\sim \text{Ber}(\gamma), \\ \sigma_\beta^2 &\sim \text{IG}(a_{\sigma_\beta}, b_{\sigma_\beta}) \end{aligned}$$

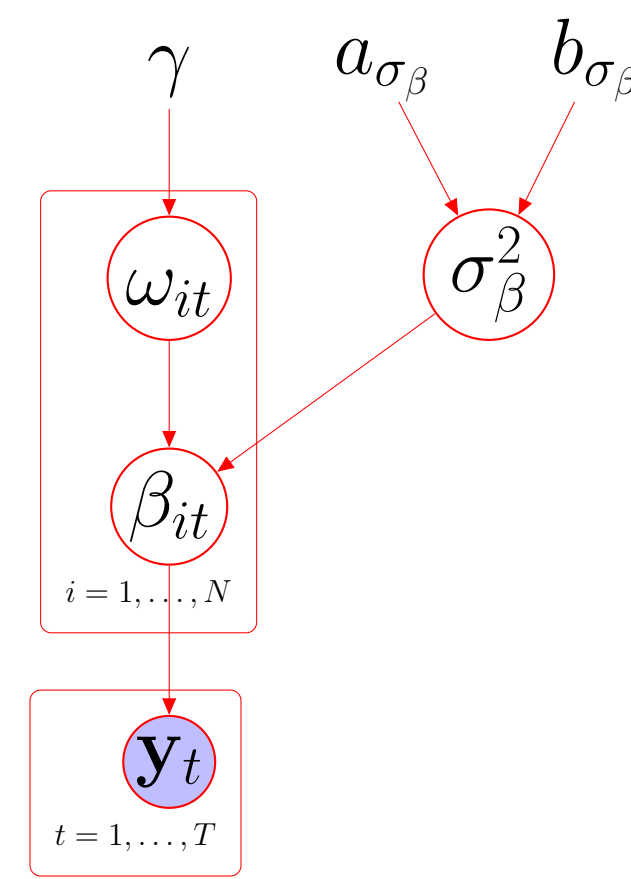


Fig. 2: Original spike and slab model

Gaussian processes

A way to specify a prior on functions.

A Gaussian process is a collection of random variables, any finite number of which have a joint Gaussian distribution.

GP prior on Bernoulli-distributed parameters allows to impose structural assumptions.

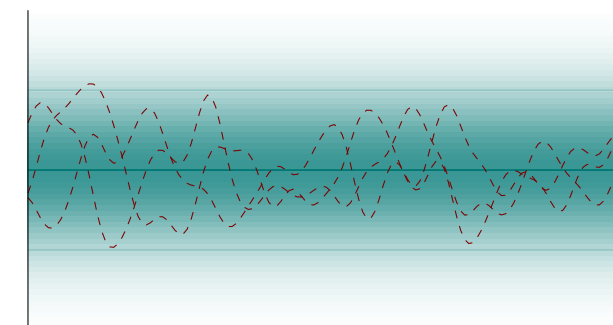


Fig. 3: Samples from GP

Proposed model

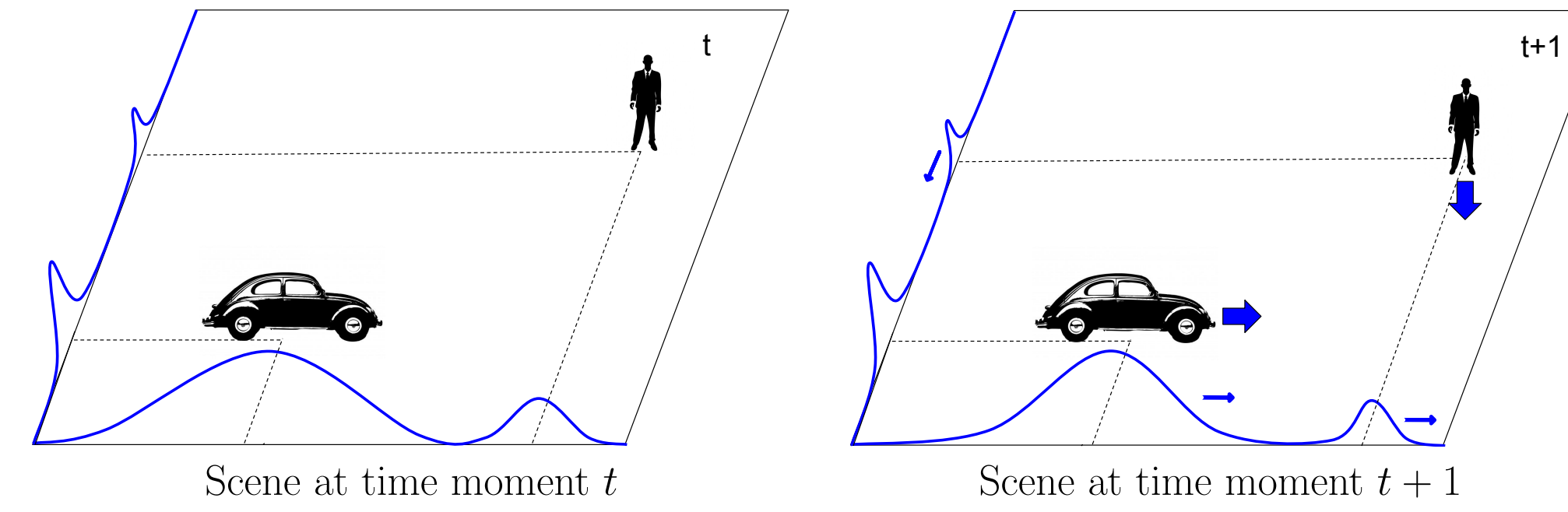


Fig. 4: Motivation example

- Extension of the spike and slab model by imposing a hierarchical Gaussian process (GP) prior
- Modelling spatial structural dependencies for coefficients that can evolve in time

Generative model:

$$\begin{aligned} \mathbf{y}_t &\sim g_t \stackrel{\text{def}}{=} \mathcal{N}(\mathbf{X}\boldsymbol{\beta}_t, \sigma^2 \mathbf{I}) \\ \beta_{it} &\sim f_{it} \stackrel{\text{def}}{=} \omega_{it}\delta_0 + (1 - \omega_{it})\mathcal{N}(0, \sigma_\beta^2) \\ \omega_{it} &\sim h_{it} \stackrel{\text{def}}{=} \text{Ber}(\Phi(\gamma_{it})) \\ \gamma_t &\sim r_t \stackrel{\text{def}}{=} \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_0) \\ \boldsymbol{\mu}_t &\sim u_t \stackrel{\text{def}}{=} \mathcal{N}(\boldsymbol{\mu}_{t-1}, \mathbf{W}) \end{aligned}$$

Assumptions:

1. $\boldsymbol{\beta}_t$ is sparse
2. non-zero elements in $\boldsymbol{\beta}_t$ are clustered in groups at every timestamp
3. these groups can move and evolve in time

Full posterior:

$$P = \prod_{t=1}^T \left[g_t \left(\prod_{i=1}^N f_{it} \right) \left(\prod_{i=1}^N h_{it} \right) r_t u_t \right]$$

Inference

Approximate the full posterior with:

$$Q = \prod_{t=1}^T \mathcal{N}(\boldsymbol{\beta}_t | \mathbf{m}_t, \mathbf{V}_t) \prod_{t=1}^T \prod_{i=1}^N \text{Ber}(\omega_{it} | \Phi(z_{it})) \prod_{t=1}^T \mathcal{N}(\gamma_t | \boldsymbol{\nu}_t, \mathbf{S}_t) \prod_{t=1}^T \mathcal{N}(\boldsymbol{\mu}_t | \mathbf{e}_t, \mathbf{D}_t)$$

Expectation Propagation

$$\begin{aligned} p(\mathcal{D}, \boldsymbol{\theta}) &= \prod_n f_n(\boldsymbol{\theta}), \\ p(\boldsymbol{\theta} | \mathcal{D}) &= \frac{1}{p(\mathcal{D})} \prod_n f_n(\boldsymbol{\theta}), \\ p(\mathcal{D}) &= \int \prod_n f_n(\boldsymbol{\theta}) d\boldsymbol{\theta}, \\ p(\boldsymbol{\theta} | \mathcal{D}) &\approx q(\boldsymbol{\theta}) = \frac{1}{Z} \prod_n \tilde{f}_n(\boldsymbol{\theta}) \end{aligned}$$

The minimisation of the KL divergence between distributions of the exponential family is equivalent to the *moments matching* between them.

Until convergence:

1. Choose $\tilde{f}_n(\boldsymbol{\theta})$ to refine.
2. Remove $\tilde{f}_n(\boldsymbol{\theta})$ from the posterior $q^{(n)}(\boldsymbol{\theta}) = \frac{q(\boldsymbol{\theta})}{\tilde{f}_n(\boldsymbol{\theta})}$.
3. Evaluate the new posterior by matching moments of $q^{(n)}(\boldsymbol{\theta})$ to those of $q^{(n)}(\boldsymbol{\theta})f_n(\boldsymbol{\theta})$.
4. Evaluate the new factor $\tilde{f}_n(\boldsymbol{\theta}) = \frac{q^{(n)}(\boldsymbol{\theta})}{q^{(n)}(\boldsymbol{\theta})}$.

Experimental results

Compressed background subtraction:

Estimate $\boldsymbol{\beta}_t$ based on $\mathbf{y}_t$ and $\boldsymbol{\beta}_0$:

$$\mathbf{y}_t = \mathbf{X}\boldsymbol{\beta}_t - \mathbf{X}\boldsymbol{\beta}_0$$

- $\boldsymbol{\beta}_0$ — background frame
- $\boldsymbol{\beta}_t$ — current frame
- $\mathbf{y}_t$ — compressed frame, $\text{length}(\mathbf{y}_t) \approx 40\% \times \text{length}(\boldsymbol{\beta}_t)$
- $\mathbf{X}$ — sensing matrix with i.i.d. Gaussian elements

The Convoy dataset (Warnell et al. 2014) is used.

Performance measures:

- *F-measure* — binary masks computed by thresholding true vectors $\boldsymbol{\beta}$ and posterior estimates $\hat{\boldsymbol{\beta}}$. The higher is the better.
- *NMSE* — normalised mean squared error between $\boldsymbol{\beta}$ and $\hat{\boldsymbol{\beta}}$. The lower is the better.

Qualitative comparison:

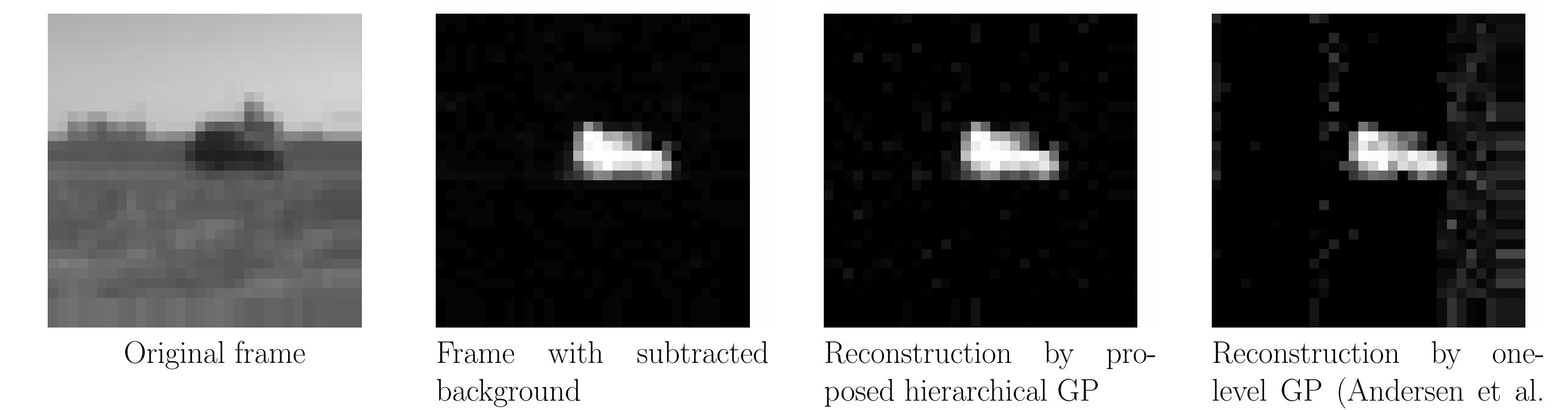


Fig. 5: Compressed sensing background subtraction performance. Sample frame with reconstruction results from sparse measurements.

Quantitative comparison:

Measure	Hierarchical GP	One-level GP
NMSE	0.0153	0.0392
F-measure	0.9870	0.9403

Tab. 1: Performance of signal reconstruction for background subtraction in video

Summary and Future work

- New model for sparse data is proposed and an efficient inference scheme is designed for it.
- Model allows to address correlations in sparse models that arise in genetics, medicine and audio/video compressive sensing.
- Model proves the possibility of combining sparse regression with a nonparametric prior, therefore it allows to implement more complex priors, for example, based on the Dirichlet process or deep Gaussian processes.